RAJALAKSHMI ENGINEERING COLLEGE

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P2

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|---|---|
| **Started on** | Monday, 31 August 2026, 5:00 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 2 September 2026, 6:58 PM |
| **Time taken** | 2 days 1 hour |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int countZeroes(int arr[], int low, int high, int n)
{
    if (low > high)
        return 0;

    int mid = (low + high) / 2;

    // If middle element is 1, search in right half
    if (arr[mid] == 1)
        return countZeroes(arr, mid + 1, high, n);

    // If mid is the first zero
    if (mid == 0 || arr[mid - 1] == 1)
        return n - mid;

    // Search in left half
    return countZeroes(arr, low, mid - 1, n);
}

int main()
{
    int m;
    scanf("%d", &m);

    int arr[m];

    for (int i = 0; i < m; i++)
        scanf("%d", &arr[i]);

    int result = countZeroes(arr, 0, m - 1, m);

    printf("%d\n", result);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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